

2025-12-12

Imkd 引发的 system_server SWT

PVT 软件部

姓名：李昌全

掌锐电子软件部

一，问题现象

monkey 测试出现 Android 黑屏

二，问题分析

2.1 解析 system_server SWT 信息

Subject: Blocked in monitor com.android.server.am.ActivityManagerService on monitor thread (watchdog.monitor) for 93s, Blocked in handler on on display thread (android.display) for 93s, Blocked in handler on on ActivityManager (ActivityManager) for 93s

watchdog.monitor 检察 AMS 状态被阻塞，其次 block display 线程和 ActivityManager

2.2 检测被 block 线程堆栈信息

1, watchdog.monitor 被 block 堆栈信息

```
1 "watchdog.monitor" prio=5 tid=12 Blocked
2   | group="main" sCount=1 ucsCount=0 flags=1 obj=0x15a003e8 self=0xb400007b38c31000
3   | sysTid=1553 nice=-2 cgrp=foreground sched=0/0 handle=0x7aa59a1cb0
4   | state=S schedstat=( 3620246794 16066202566 27490 ) utm=36 stm=325 core=0 HZ=100
5   | stack=0x7aa589e000-0x7aa58a0000 stackSize=1039KB
6   | held mutexes=
7   at com.android.server.am.ActivityManagerService.monitor(ActivityManagerService.java:17350)
8   - waiting to lock <0x0fe7e42a> (a com.android.server.am.ActivityManagerService) held by thread 16
9   at com.android.server.Watchdog$HandlerChecker.run(Watchdog.java:385)
10  at android.os.Handler.handleCallback(Handler.java:958)
11  at android.os.Handler.dispatchMessage(Handler.java:99)
12  at android.os.Looper.loopOnce(Looper.java:219)
13  at android.os.Looper.loop(Looper.java:316)
14  at android.os.HandlerThread.run(HandlerThread.java:67)
15  at com.android.server.ServiceThread.run(ServiceThread.java:46)
16
```

2, display thread 被 block 堆栈信息

```
"android.display" prio=5 tid=16 Native
| group="main" sCount=1 ucsCount=0 flags=1 obj=0x15a00768 self=0xb400007b38c36400
| sysTid=1558 nice=-3 cgrp=top-app sched=0/0 handle=0x7aa1579cb0
| state=S schedstat=( 2610294190406 7248631711867 15356939 ) utm=113764 stm=147264 core=6 HZ=100
| stack=0x7aa1476000-0x7aa1478000 stackSize=1039KB
| held mutexes=
native: #00 pc 000ed744 /apex/com.android.runtime/lib64/bionic/libc.so (__sendmsg+4) (BuildId: 2bfa364faa6bccdd18da15e2922c1c52)
native: #01 pc 0001476c /system/lib64/libbase.so (android::base::SendFileDescriptorVector+524) (BuildId: ab094e01788278cfd782ea8237c7ffee)
native: #02 pc 0018efac /system/lib64/libandroid_runtime.so (android::socket_write_all+380) (BuildId: 3df117c62050d93a1923b2ecf07a00af)
native: #03 pc 0018e588 /system/lib64/libandroid_runtime.so (android::socket_writeba+424) (BuildId: 3df117c62050d93a1923b2ecf07a00af)
at android.net.LocalSocketImpl.writeba_native(Native method)
at android.net.LocalSocketImpl.$Nest$writeba_native(unavailable:0)
at android.net.LocalSocketImpl$SocketOutputStream.write(LocalSocketImpl.java:144)
- locked <0x073b0381> (a java.lang.Object)
at com.android.server.am.LmkdConnection.write(LmkdConnection.java:267)
- locked <0x05f7fc26> (a java.lang.Object)
at com.android.server.am.LmkdConnection.exchange(LmkdConnection.java:293)
at com.android.server.am.ProcessList.writeLmkd(ProcessList.java:1602)
at com.android.server.am.ProcessList.setOomAdj(ProcessList.java:1517)
at com.android.server.am.OomAdjuster.applyOomAdjLSP(OomAdjuster.java:2980)
at com.android.server.am.OomAdjuster.updateAndTrimProcessLSP(OomAdjuster.java:1299)
at com.android.server.am.OomAdjuster.updateOomAdjInnerLSP(OomAdjuster.java:1051)
at com.android.server.am.OomAdjuster.performUpdateOomAdjLSP(OomAdjuster.java:595)
at com.android.server.am.OomAdjuster.updateOomAdjLSP(OomAdjuster.java:581)
at com.android.server.am.OomAdjuster.updateOomAdjLocked(OomAdjuster.java:569)
- locked <0x0ebc33a7> (a com.android.server.am.ActivityManagerProcLock)
at com.android.server.am.ActivityManagerService.updateOomAdjLocked(ActivityManagerService.java:16809)
at com.android.server.am.ActivityManagerService.trimApplicationsLocked(ActivityManagerService.java:17108)
at com.android.server.am.ActivityManagerService.trimApplications(ActivityManagerService.java:17060)
- locked <0x0fe7e42a> (a com.android.server.am.ActivityManagerService)
at com.android.server.am.ActivityManagerService.$Nest$mtrimApplications(ActivityManagerService.java:0)
at com.android.server.am.ActivityManagerService$LocalService.trimApplications(ActivityManagerService.java:18185)
at com.android.server.wm.ActivityTaskSupervisor.lambda$activityIdleInternal$2(ActivityTaskSupervisor.java:1511)
at com.android.server.wm.ActivityTaskSupervisor.$r8$lambda$TrgvD2zUzuQJBT9Ga5MwmE2w2s(ActivityTaskSupervisor.java:0)
at com.android.server.wm.ActivityTaskSupervisor$ExternalSyntheticLambda9.run(R8$$SyntheticClass:0)
at android.os.Handler.handleCallback(Handler.java:958)
at android.os.Handler.dispatchMessage(Handler.java:99)
at android.os.Looper.loopOnce(Looper.java:219)
at android.os.Looper.loop(Looper.java:316)
at android.os.HandlerThread.run(HandlerThread.java:67)
at com.android.server.ServiceThread.run(ServiceThread.java:46)

"CpuTracker" prio=5 tid=35 TimedWaiting
```

3, ActivityManager 被 block 堆栈信息

```
41 "ActivityManager" prio=5 tid=25 Blocked
42 | group="main" sCount=1 ucsCount=0 flags=1 obj=0x15a00ef0 self=0xb400007b38bd2c00
43 | sysTid=1611 nice=-2 cgrp=foreground sched=0/0 handle=0x7a979a7cb0
44 | state=S schedstat=( 478374812711 2858948462666 3333580 ) utm=21986 stm=25850 core=7 HZ=100
45 | stack=0x7a978a4000-0x7a978a6000 stackSize=1039KB
46 | held mutexes=
47 at com.android.server.am.BroadcastQueueModernImpl.lambda$new$0(BroadcastQueueModernImpl.java:295)
48 - waiting to lock <0x0fe7e42a> (a com.android.server.am.ActivityManagerService) held by thread 16
49 at com.android.server.am.BroadcastQueueModernImpl.$r8$lambda$EZvetGiyPFSQ7Dv6AGCYMZgJrJo(BroadcastQueueModernImpl.java:0)
50 at com.android.server.am.BroadcastQueueModernImpl$ExternalSyntheticLambda7.handleMessage(R8$$SyntheticClass:0)
51 at android.os.Handler.dispatchMessage(Handler.java:102)
52 at android.os.Looper.loopOnce(Looper.java:219)
53 at android.os.Looper.loop(Looper.java:316)
54 at android.os.HandlerThread.run(HandlerThread.java:67)
55 at com.android.server.ServiceThread.run(ServiceThread.java:46)
```

从三个线程被block的堆栈信息看,根因在AMS updateOomAdjLocked()更新oomadj值到lmkd被阻塞在socket传输上,lmkd存在异常

2.3 查询lmkd状态

lmkd 日志分析

06-23 04:37:55.263	349	362	E	lowmemorykiller:	epoll_wait	failed	(errno=22)
06-23 04:37:55.263	349	362	E	lowmemorykiller:	epoll_wait	failed	(errno=22)
06-23 04:37:55.263	349	362	E	lowmemorykiller:	epoll_wait	failed	(errno=22)
06-23 04:37:55.263	349	362	E	lowmemorykiller:	epoll_wait	failed	(errno=22)
06-23 04:37:55.263	349	362	E	lowmemorykiller:	epoll_wait	failed	(errno=22)
06-23 04:37:55.263	349	362	E	lowmemorykiller:	epoll_wait	failed	(errno=22)
06-23 04:37:55.263	349	362	E	lowmemorykiller:	epoll_wait	failed	(errno=22)
06-23 04:37:55.263	349	362	E	lowmemorykiller:	epoll_wait	failed	(errno=22)
06-23 04:37:55.263	349	362	E	lowmemorykiller:	epoll_wait	failed	(errno=22)
06-23 04:37:55.263	349	362	E	lowmemorykiller:	epoll_wait	failed	(errno=22)
06-23 04:37:55.263	349	362	E	lowmemorykiller:	epoll_wait	failed	(errno=22)
06-23 04:37:55.263	349	362	E	lowmemorykiller:	epoll_wait	failed	(errno=22)
06-23 04:37:55.263	349	362	E	lowmemorykiller:	epoll_wait	failed	(errno=22)
06-23 04:37:55.263	349	362	E	lowmemorykiller:	epoll_wait	failed	(errno=22)
06-23 04:37:55.263	349	362	E	lowmemorykiller:	epoll_wait	failed	(errno=22)
06-23 04:37:55.263	349	362	E	lowmemorykiller:	epoll_wait	failed	(errno=22)
06-23 04:37:55.263	349	362	E	lowmemorykiller:	epoll_wait	failed	(errno=22)
06-23 04:37:55.263	349	362	E	lowmemorykiller:	epoll_wait	failed	(errno=22)
06-23 04:37:55.263	349	362	E	lowmemorykiller:	epoll_wait	failed	(errno=22)
06-23 04:37:55.263	349	362	E	lowmemorykiller:	epoll_wait	failed	(errno=22)
06-23 04:37:55.263	349	362	E	lowmemorykiller:	epoll_wait	failed	(errno=22)

```
I HealthMain_CPUMonitor: 800%cpu 78%user 0%nice 320%sys 243%idle 140%iow 17%irq 2%sirq 0%host
I HealthMain_CPUMonitor: PID USER PR NI VIRT RES SHR S[%CPU] %MEM TIME+ ARGS
I HealthMain CPUMonitor: 349 lmkd -2 0 2.3G 5.6M 4.5M R 273 0.0 15:36.75 lmkd
```

lmkd cpuload 异常高，其次存在非常多 epoll_wait 异常打印。

2.4 lmkd code 排查

thread 362
thread 363

结论：第一次出现问题的是 thread 362，监控 epoll cpu fd

Commit	Author	Date	Message	
2025/4/11	shizongxuan	4799	4810	#endif
2025/4/11	shizongxuan	4800	4811	
2025/5/15	Detaoyu	4801	4812	std::thread psi_cpu_thread(mainloop, epoll_cpu_fd);
2025/5/15	Detaoyu	4802	4813	psi_cpu_thread.detach();
2025/5/15	Detaoyu	4803	4814	结论：1. 此出错非原生的Android lmkd 代码
2025/5/15	Detaoyu	4804	4815	2. 此段代码由 Detaoyu@nuoboauto.com 与 2025/5/15 提交的代码
2025/5/15	Detaoyu	4806	4817	
2025/5/15	Detaoyu	4807	4818	mainloop(epollfd);
2018/4/3	Mark	4808	4819	}
2013/8/31	Todd	4809	4820	
2018/8/11	Suren	4810	4821	android_log_destroy(&ctx);
2018/8/11	Suren	4811	4822	
2013/8/31	Todd	4812	4823	ALOGI("exiting");
2013/8/31	Todd	4813	4824	return 0;
2025/5/15	Detaoyu	4814	4825	

Commit aee25a5f92940e47196f895ba8b300bd0dfded0
Author: Detaoyu
Date: 2025/5/15

[None]移植lmkd中对cpu io的监听代码到V5平台

[MODULE]:lmkd
[REASON]:CR
[SOLUTION]:代码实现
[IMPACT]:no
[TEST RECOMMENDATIONS]:代码实现

Change-Id: I79cbf13ea4b0453fc4c09edfdade821c820d1

三、问题结论

lmkd 存在大量修改，经沟通，存在客户修改，是从单系统项目迁移过来，而多系统 lmkd mtk 存在已客制化，客户从单系统移值不适用。

四、解决方案

回退 客户从单系统移值 lmkd 修改。